Sharzil Hasnine

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Portfolio: shas-9.github.io

Education

Bachelor of Computer Science (Advanced) Feb 2023 – Nov 2026
Adelaide University

Technical Skills

Languages: C++, Java, TypeScript, JavaScript, Python, C, GDScript, MATLAB, Bash
Libraries and Frameworks: Node.js, Spring Boot, MongoDB, MySQL, React, Flask, JUnit, Godot, Pandas, NumPy, Matplotlib, Sklearn, TensorFlow, ROS2
Tools and Software: Git, Linux, Docker, Jira, Gazebo, Nav2

Experience

Student Data Engineer - REDCap-KNIME Integration - Adelaide University Aug 2026 - Oct 2026
Tech: *Python, KNIME, REDCap*

- Connected a research data system (REDCap) to an analytics workflow tool (KNIME) by building a custom Python extension using PyCap to pull records and metadata through the REST API into separate typed output tables.
- Implemented metadata-driven ETL and validation: decoded categorical and checkbox fields, enforced field-level validation rules, and preserved longitudinal event and repeating-instrument context.
- Built a controlled test environment of 10,000 synthetic records across 64 fields (no real participant data), enabling reproducible validation of the extraction and ETL workflow.

Industry Project - Block Model Compression Algorithm - Maptek Aug 2025 - Oct 2025
Tech: *C++, Bash, Jira*

- Architected a stream-oriented lossless compression system for 3D data, optimising for speed and compression ratio.
- Implemented an octree-based spatial compression algorithm, achieving 98.8% reduction in data size while improving processing speed by 2.6x over the uncompressed baseline, forming the foundation for subsequent team optimisations reaching 99.17% compression and 1520x processing speed.
- Designed a testing framework that verifies correctness of compressed outputs against uncompressed input data, ensuring algorithmic correctness independent of compression strategy and enabling automated, extensible testing.
- Built a cross-platform CI pipeline with automated testing and reproducible Linux and Windows builds, integrating cross-compilation and artifact generation into the development workflow, improving team velocity by 1.5x.

Software Developer - Autonomous Navigation - Adelaide Rover Team Oct 2024 - Feb 2025
Tech: *Python, C++, ROS2, Nav2, Gazebo, Groot*

- Designed a behavior tree for the rover's autonomous navigation pipeline using Nav2, enabling modular decision-making for tasks such as path planning, recovery behaviours, and goal execution.
- Simulated navigation behaviour using Gazebo, supporting the Adelaide Rover Team's preparation for the Australian Rover Challenge's autonomous navigation task.

Projects

Daily Iftar Registration System Feb 2026 - Mar 2026
Tech: *TypeScript, Node.js, MongoDB, REST API, Docker*
Link: *aumsa.com.au*

- Delivered production-ready backend features under tight release cycles, building and iterating on RESTful APIs (Node.js, MongoDB) to support live event registration used by 70+ daily users.
- Architected DST-safe, timezone-aware date handling (Australia/Adelaide), eliminating off-by-one scheduling bugs and ensuring reliable, real-world consistency for day-based operations.
- Designed scalable MongoDB database schemas and API contracts to support rapid feature expansion and seamless frontend integration, maintaining stability while continuously deploying new functionality.

Paxos Consensus Algorithm Sep 2025 - Oct 2025
Tech: *Java, JUnit, Bash*
Link: *github.com/Shas-9/Paxos*

- Implemented the Paxos Consensus Algorithm without use of any external libraries, enabling multiple concurrent nodes to elect a single leader despite network delays and node failures.
- Developed a TCP socket-based messaging protocol supporting Paxos phases (prepare, promise, accept, learn) with nodes dynamically acting as proposers, acceptors, and learners to achieve consensus.
- Designed a failure simulation framework introducing latency, message loss, and node crashes to validate consensus correctness under concurrent proposals and partial system failures.

Distributed Weather Feed Aggregator with Lamport Clocks Aug 2025 - Sep 2025

Tech: *Java, JUnit, REST API*

Link: *github.com/Shas-9/Weather-Aggregation-Server*

- Engineered a distributed data aggregation system supporting concurrent clients and content servers.
- Implemented Lamport logical clocks to maintain causal consistency and deterministic ordering of concurrent updates across distributed components, without use of external libraries.
- Developed a multi-threaded aggregation server with thread-safe request processing, persistent storage with crash recovery and automated 30s expiry of stale data sources, ensuring system reliability under concurrent workloads.

SVM and Neural Network Implementation Aug 2025 - Oct 2025

Tech: *Python, NumPy, Sklearn*

Link: *github.com/Shas-9/SVM-and-Neural-Network-Implementation*

- Implemented a Support Vector Machine from scratch in Python by solving the dual optimization problem with quadratic programming, including RBF kernel support and hyperparameter tuning using validation data.
- Built a two-layer neural network from scratch using forward and backward propagation, implementing core modules (Linear, ReLU, Cross-Entropy) and training with mini-batch SGD on a dataset with 200 features and 8.5k samples.
- Developed an end-to-end ML training pipeline including data standardization, train/validation/test splits, hyperparameter search, and performance evaluation using accuracy metrics with Sklearn.

CatQuest – 2D Maze Exploration Game Aug 2023 - Oct 2023

Tech: *C++, SFML*

Link: *github.com/Shas-9/Labyrinth-game*

- Engineered modular game architecture, separating game state, entities, and UI for maintainability and scalability.
- Integrated enemy AI and player mechanics within the existing procedural random maze generation system, including real-time collision handling and health management.
- Designed the game UI, with seamless navigation and state transitions without affecting game loop performance.

Monkey Run – Unihack 2024 Mar 2024

Tech: *GDScript, Godot, Python, MediaPipe*

Link: *github.com/rayokamoto/unihack*

- Developed an endless runner game with randomised object generation using Godot game engine with GDScript.
- Implemented input detection by tracking player movement through webcam by using MediaPipe ML framework.
- Integrated input detection system with game mechanics to deliver a cohesive gaming experience.

Extracurriculars

Adelaide University Competitive Programming League (AUCPL) Mar 2026 – Oct 2026

- Achieved a team 2nd place finish in the Adelaide University Competitive Programming League.

Secretary Nov 2024 – Aug 2026

ISSUA

- Led internal communications for a 10+ member committee, producing structured meeting minutes and action item documentation to ensure organizational alignment and accountability.
- Managed university-facing logistics including room bookings, building access, and prayer room maintenance.
- Administered organizational knowledge management using Google Drive, maintaining structured repositories of meeting records, event documentation, and operational resources to support cross-functional collaboration.

Events Officer Nov 2023 – Nov 2024

ISSUA

- Planned and executed society events for 100+ attendees, overseeing end-to-end logistics, budget management, and stakeholder alignment to meet organizational objectives.
- Coordinated event setup, including venue preparation, infrastructure, and audio-visual systems.
- Engaged with attendees and guests during events, providing support and fostering a welcoming environment.